

Burkitt lymphoma

MYC-driven, Ki-67 ~100%, curable — if you prevent tumor lysis, cover the CNS, and give intensive rituximab-containing therapy without delay. The regimen decision turns on **age, fitness, and whether the CSF is involved**

Compiled from the *Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin* — *Burkitt and HIV-associated lymphoma* · NCCN · primary trial reports verified against PubMed, September 2026.

BIOLOGY AND DIAGNOSIS

Defining lesion	MYC rearrangement to an Ig locus, confirmed by FISH: t(8;14) IGH ~80% · t(2;8) IGH ~15% · t(8;22) IGL ~5%. Co-operating ID3, TCF3, TP53 mutations
Morphology	Monomorphic medium cells, diffuse. “ Starry sky ” — tingible-body macrophages engulfing apoptotic debris
Phenotype	CD20+, CD10+ , BCL6+ , BCL2 negative , TdT negative · Ki-67 ~100%
Endemic	Equatorial Africa, children; EBV+ in nearly all , malaria co-factor. Jaw / orbit , abdominal viscera
Sporadic	Worldwide, all ages; EBV+ in a minority. Ileocecal / abdominal mass , marrow, CNS
Immunodeficiency	HIV (often at relatively preserved CD4), post-transplant. Nodal, marrow and CNS disease common
Not Burkitt	BCL2 or BCL6 rearranged too → HGBCL double-hit (treat per DLBCL card). MYC-negative with 11q gain/loss → HGBCL with 11q aberration (WHO-HAEM5; ICC: LBCL with 11q), formerly “Burkitt-like”. TdT+ → lymphoblastic

WHO-HAEM5 now subtypes Burkitt by **EBV status** (EBV+ vs EBV–) rather than by the epidemiologic variant.

WORKUP — TREAT AS AN EMERGENCY

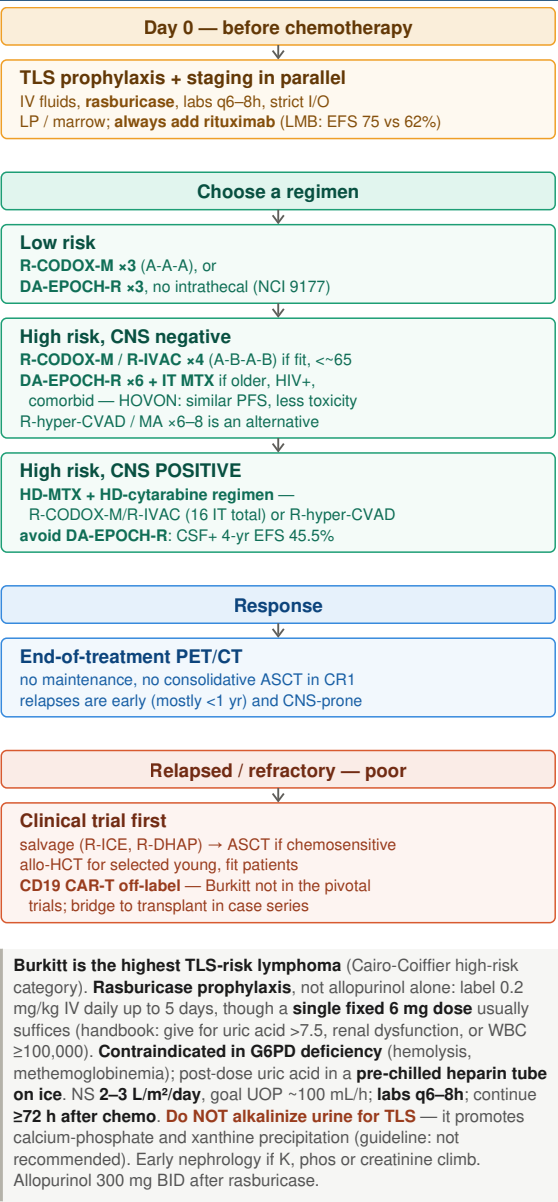
- **Excisional or core biopsy with MYC FISH** (plus BCL2, BCL6 FISH); start therapy within days
- **TLS labs at presentation**: K, phos, Ca, uric acid, creatinine, **LDH** — spontaneous TLS before any chemo is classic
- **PET/CT · bone marrow biopsy · LP with cytology AND flow cytometry** (brain MRI if symptoms) — CNS involved in ~**19%** of adults
- **HIV**, HBsAg / anti-HBc / anti-HBs, HCV · **echo** · **G6PD before rasburicase** · pregnancy test · fertility counseling
- **No LP if circulating blasts or platelets <50,000**

RISK STRATIFICATION

Low risk (regimen gate)	All of: normal LDH, ECOG 0–1, stage I–II, no mass ≥10 cm, <1 extranodal site (Magrath / Mead). NCI 9177 used mass <7 cm , not 10. Everyone else is high risk (~85–90% of adults)
BL-IPI (prognosis)	1 point each: age ≥40 · ECOG ≥2 · LDH >3× ULN · CNS involvement . 3-yr PFS 0 = 92% · 1 = 72% · ≥2 = 53% <i>Olszewski, JCO 2021</i>

Real-world (30 US centres, 641 adults): 3-yr PFS **64%**, **treatment-related mortality 10%** — well below trial results. Rituximab (PFS 67% vs 38%) and **academic-centre care** (67% vs 46%) tracked with better outcome
Evens, Blood 2021.

TREATMENT PATHWAY



REGIMEN CHOICE — WHAT THE EVIDENCE SAYS

Regimen	Evidence	Weak spot
R-CODOX-M / R-IVAC <i>modified Magrath</i>	LY10 (Mead 2008, MTX cut to 3 g/m ²): 2-yr PFS 85% low risk, 49% high risk , with less toxicity than full-dose. Lowest CNS relapse in real-world data (~4%)	Most toxic: grade 3–5 infection 56% (HOVON). Inpatient; age >65 needs dose reduction
DA-EPOCH-R	NCI 9177 (single arm, n=113, 25% HIV+): 4-yr EFS 84.5% — 100% low risk, 82% high risk; same across age and HIV. HOVON/SAKK randomized (CNS-negative high risk): 2-yr PFS 70% vs 76%, HR 1.42, p=0.40 — underpowered, closed early	CNS . CSF+ high-risk EFS 45.5% vs 89.9% . Real-world CNS relapse 13% vs 3–4% with MTX/cytarabine regimens (aSHR 4.38)
R-hyper-CVAD / MA	MDACC, n=31 (29% ≥60 yr): CR 86%, 3-yr OS 89% , EFS 80%. Single arm	Long (6–8 cycles), myelosuppressive; small series
Rituximab	Ribrag, Lancet 2016 (LMB ± R, n=260, randomized): 3-yr EFS 75% vs 62% , HR 0.59, no added toxicity	— add to every regimen

No regimen has beaten another for survival in a completed randomized trial. Practical rule: **fit, younger, or CNS-positive** → **HD-MTX/cytarabine-based**; **older, HIV+, or frail and CNS-negative** → **DA-EPOCH-R**.

CNS — EVERYONE GETS COVERED

- **Prophylaxis is universal** (IT MTX ± cytarabine built into every regimen, + systemic HD-MTX in Magrath and hyper-CVAD). The one exception: **NCI 9177 low-risk patients received none** and none relapsed
- **CSF involvement is the strongest adverse factor** in every series — stage it properly (cytology + flow) before cycle 1 IT
- **CNS-positive**: intensify IT (Magrath: 2 extra per cycle, 16 total; hyper-CVAD: IT twice weekly until CSF clear ×2) — systemic HD-MTX and HD-cytarabine penetrate the parenchyma, infusional EPOCH does not

HIV-ASSOCIATED BURKITT

- **Treat with curative intent, same regimens** — HIV status no longer predicts outcome (NCI 9177 EFS 84.9% HIV+ vs 84.5% HIV–; US-UK series 3-yr PFS 61%)
- **Continue or start ART** with oncology-pharmacy review; **avoid ritonavir / cobicistat boosting** (CYP3A4 → vincristine neuropathy, anthracycline and etoposide toxicity) — an integrase-inhibitor backbone is usual
- US-UK HIV-BL cohort (n=249): baseline CNS disease 3-yr PFS 36% vs 69%; **CNS recurrence higher after DA-EPOCH** (sHR 2.52); CODOX-M/IVAC associated with longer PFS (aHR 0.45) — retrospective
- PJP prophylaxis throughout; **MAC prophylaxis if CD4 <100**; consider quinolone during nadir

Burkitt lymphoma

MODIFIED MAGRATH (MEAD, BLOOD 2008) — Q21D OR ON COUNT RECOVERY

Cycle	Dosing (mg/m² unless stated)
A — R-CODOX-M	Rituximab 375 d1 · cyclophosphamide 800 d1 + 200 d2–5 (first A; later A cycles 800 d1+2) · doxorubicin 40 d1 · vincristine 1.5 (cap 2 mg) d1+8 · MTX 3000 over 24 h on d10 (300 over 1 h, then 2700 over 23 h) · IT cytarabine 70 mg d1+3 · IT MTX 12 mg d15
B — R-IVAC	Rituximab 375 d1 · etoposide 60 d1–5 · ifosfamide 1500 d1–5 + mesna · cytarabine 2000 q12h x4 d1–2 (steroid eye drops) · IT MTX 12 mg d5
Age >65	MTX 1000 (100 over 1 h, then 900 over 23 h) · ifosfamide 1000 · cytarabine 1000
CNS positive	A: + IT cytarabine 70 mg d5, IT MTX 12 mg d17 · B: + IT cytarabine 70 mg d7 and d9 — 16 IT total
Leucovorin	15 mg/m² IV at 36 h from MTX start, q3h to 48 h, then q6h until MTX <0.05 µM. After each IT MTX: 15 mg PO at 24 h
Start cycle if	ANC >1000, plt >75k, direct bili <1.4, Cr <1.2 — not required for cycle 1 or for d10 HD-MTX

Rituximab was not part of LY10 (added in routine practice; Ribrag). G-CSF after the MTX level is <0.02 (A); d7–16 or pegfilgrastim d7 (B). Levofloxacin during nadir.

DA-EPOCH-R (ROSCHESKI, JCO 2020) — Q21D

	Dosing
Drugs	Rituximab 375 d1 · etoposide 50, doxorubicin 10, vincristine 0.4 (no cap) , each /day CIV d1–4 · cyclophosphamide 750 d5 · prednisone 60 BID PO d1–5 (handbook lists 60/day)
Cycles	Low risk x3, no IT · high risk x6 + IT MTX 12 mg each cycle (extended IT if CSF+)
Adjust	CBC twice weekly; ANC nadir ≥500 → +20% (E, D, C) · <500 on 1–2 checks → same · <500 on 3 checks or plt <25k → –20%

G-CSF from 24 h after chemo. Outpatient-feasible via ambulatory pump.

SUPPORTIVE CARE THAT TRIPS PEOPLE UP

- **Hold TMP-SMX around HD-MTX** (delays clearance) — use atovaquone or pentamidine for PJP, or resume once MTX has cleared
- **Avoid azoles that inhibit CYP3A4 (posaconazole, voriconazole, itraconazole) with vincristine** — severe neuropathy and ileus. Bowel regimen with every vincristine dose
- **HD-cytarabine**: prednisolone 1% drops 2 gtt OU BID from 12 h before to 48 h after; **check for cerebellar signs before each dose** (higher risk with age >60 or renal dysfunction)
- **Ifosfamide**: mesna with every dose; watch for encephalopathy
- **HBV**: entecavir if HBsAg+ or anti-HBc+ on rituximab · G-CSF with every cycle · quinolone during neutropenia

R-HYPER-CVAD / R-MA (THOMAS, CANCER 2006) — ALTERNATING, 6–8 CYCLES

	Dosing
A (1,3,5,7)	Rituximab 375 · cyclophosphamide 300 q12h x6 d1–3 (mesna) · doxorubicin 50 CIV d4 · vincristine 2 mg d4+11 · dexamethasone 40 mg d1–4, 11–14
B (2,4,6,8)	Rituximab 375 · MTX 200 over 2 h + 800 over 22 h d1 · cytarabine 3000 q12h x4 d2–3 (1000 if >60 yr or Cr >1.5) · leucovorin from 36 h
CNS	IT MTX 12 mg d2 + IT cytarabine 100 mg d8, cycles 1–4 (8 IT). CNS+: twice weekly until CSF clear x2

HIGH-DOSE MTX — THE ALKALINIZATION THAT DOES APPLY

Pre-check	Drain or avoid if effusion / ascites (third-space reservoir) · stop PPIs, TMP-SMX, penicillins, NSAIDs, salicylates, probenecid · CrCl: 50% dose if 10–50, omit <10
Alkalinize	½NS + 100 mEq NaHCO₃/L at 500 mL/h x2 h, then 125 mL/h. Start MTX only when urine pH ≥7 and UOP >100 mL/h; keep pH ≥7 until cleared
Levels	q24h from 24 h. Escalate leucovorin to 100 mg/m² q6h if at 24 h Cr ↑>50% or level >20 µM; 48 h >1 µM; 72 h >0.1 µM
Glucarpidase	50 U/kg IV if level >1 µM with delayed clearance from renal failure. No leucovorin within 2 h either side; immunoassay unreliable for 48 h (use chromatographic assay)

Contrast with TLS (page 1): urine alkalinization is **required for HD-MTX** (MTX precipitates in acid urine) but **harmful for TLS**. When both overlap (cycle 1 d10), rasburicase has already handled urate — alkalinize for the MTX.

KEY TRIALS AND COHORTS

LY10 <i>Mead, Blood 2008</i>	Dose-modified CODOX-M/IVAC (MTX 3 g/m²), prospective: 2-yr PFS 64% in BL; low risk 85%, high risk 49%
LMB ± R <i>Ribrag, Lancet 2016</i>	Phase 3, n=260, HIV–: rituximab 3-yr EFS 75% vs 62% , HR 0.59 — the rituximab randomization
NCI 9177 <i>Roschewski, JCO 2020</i>	Risk-adapted DA-EPOCH-R, single arm: EFS 84.5%; CSF+ is the failure group
HOVON/SAKK <i>Chamuleau, Lancet Haem 2023</i>	R-CODOX-M/R-IVAC vs DA-EPOCH-R, CNS– high risk: 2-yr PFS 76% vs 70%, not significant ; closed early (n=89); fewer infections with EPOCH
MDACC R-hyper-CVAD <i>Thomas, Cancer 2006</i>	n=31: 3-yr OS 89%; rituximab improved outcome vs historical hyper-CVAD
BL-IPi <i>Olszewski, JCO 2021</i>	Age ≥40, ECOG ≥2, LDH >3x, CNS — 3-yr PFS 92 / 72 / 53%; validated in 457 ex-US patients
US real-world <i>Evans, Blood 2021</i>	641 adults, 30 centres: 3-yr PFS 64%, OS 70%, TRM 10% ; no difference by HIV status
CNS cohort <i>Zayac, Haematologica 2021</i>	CNS+ in 19%; CR 59% vs 77%. 3-yr CNS relapse 13% after DA-EPOCH-R vs 4% CODOX-M/IVAC, 3% hyper-CVAD/MA
HIV-BL <i>Alderuccio, Blood Adv 2021</i>	n=249 US-UK: 3-yr PFS 61%; CD4 did not predict; CNS relapse higher after DA-EPOCH

PEARLS

- **Board triad: MYC-IG translocation + BCL2-negative + Ki-67 ~100%**. BCL2-positive with MYC = think double hit, not Burkitt
- **Urine alkalinization: never for TLS, always for HD-MTX**
- Don't wait for complete staging to start TLS prophylaxis — several pediatric-derived protocols (LMB) open with a low-intensity **pre-phase** (cyclophosphamide, vincristine, prednisone) to debulk
- **If CSF is positive, don't choose DA-EPOCH-R**; if the patient is old, HIV+ or frail and CSF-negative, DA-EPOCH-R is the kinder, equally effective choice
- No maintenance, no consolidative transplant in CR1